

Lecture 6

Measures of Spread

Review

1 One value is far larger than all the rest. Which moves more, the mean or the median?

The mean, which gets pulled toward it. The median barely moves – it only cares about position in the ordered list.

2 We wrote the mean three different ways last time. What were they?

Add the values and divide by n ; or work from a frequency table; or from a relative frequency table. All three give the same number.

3 A student's exam score is at the 75th percentile. What does that tell you?

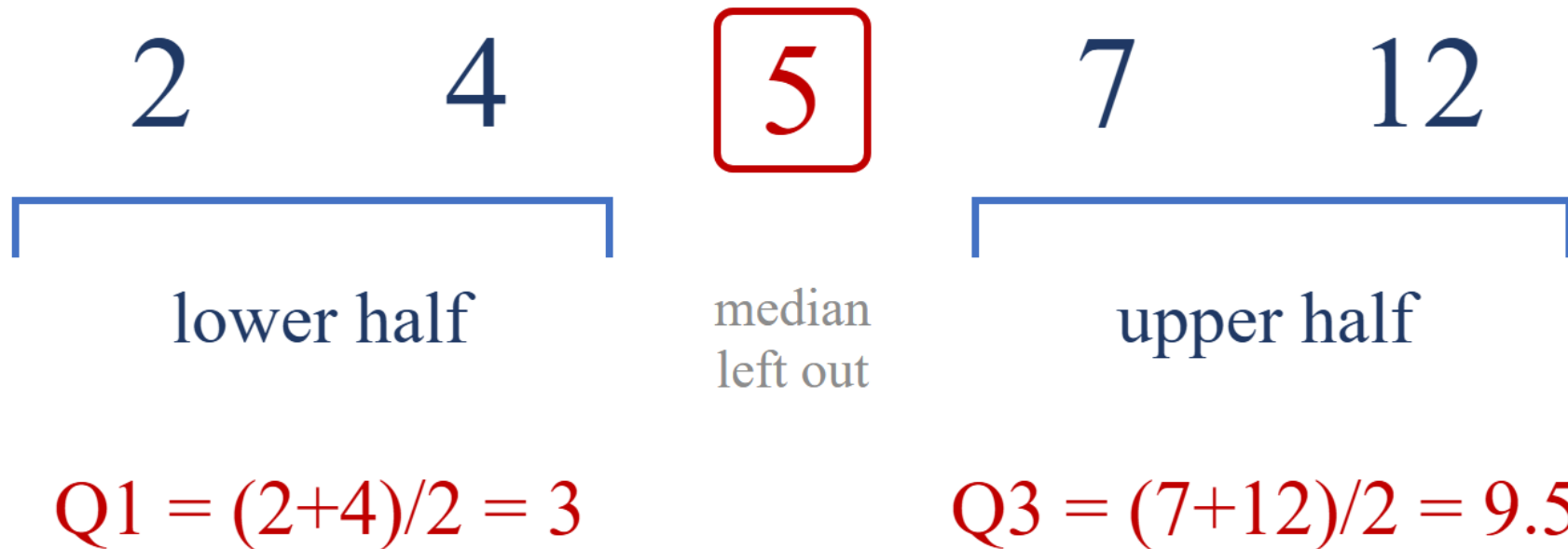
That 75 percent of the scores are at or below theirs.

4 Once you have the median, how do you find Q1 and Q3?

Q1 is the median of the lower half and Q3 the median of the upper half, leaving the median itself out of both halves when n is odd.

A note on quartiles: which definition?

- There are several accepted ways to calculate a quartile. In this class we use the **median of halves** method.
- Find the median first. When n is odd, leave that median out when forming the lower and upper halves.



Why R may give you different quartiles

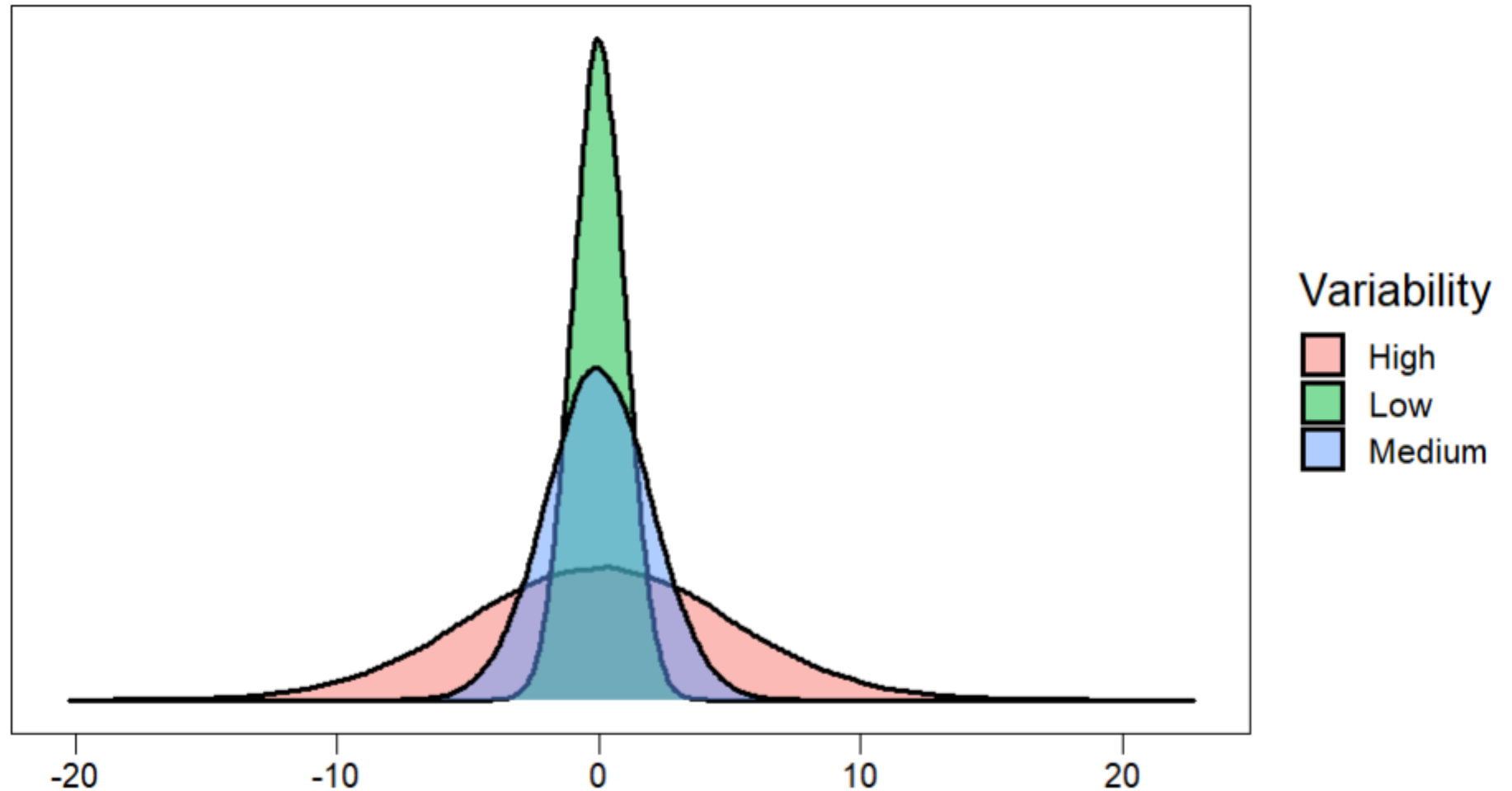
Method	Q1	Median	Q3
Median of halves – this class	3	5	9.5
R quantile() default	4	5	7
R boxplot() hinges	4	5	7

- R's quantile() interpolates between the ordered values; R's boxplot() is different again, drawing the box at Tukey's hinges, which keep the median inside both halves
- A disagreement does not mean either of you made a mistake - it means a different definition was used
- In this class, use the median of halves, and apply the same method **consistently**

Our running example for the rest of today

- We will build every measure of spread from the same five values, on the board
 - $x = 2, 4, 5, 7, 12$ $n = 5$
 - $\text{mean} = (2 + 4 + 5 + 7 + 12) / 5 = 30 / 5 = 6$
- and we already have its quartiles, from the slide before:
 - $Q1 = 3$ $Q2 = 5$ $Q3 = 9.5$
- Still to come: the range, the IQR, the variance and standard deviation, and the boxplot
- Keep these numbers - everything from here on is worked from them

Variability of A Distribution: Measures of Spread

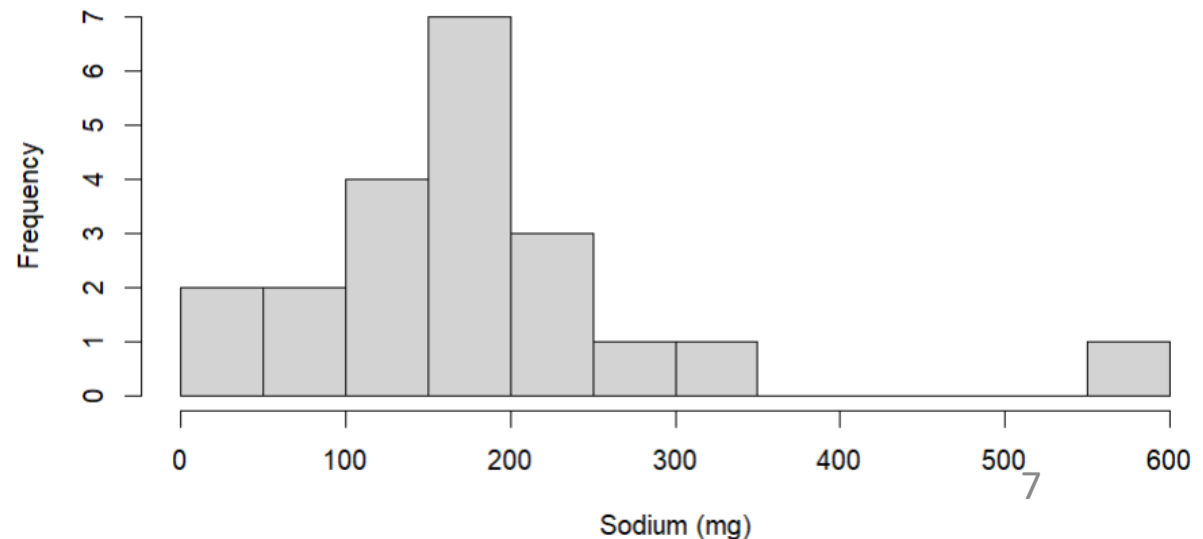
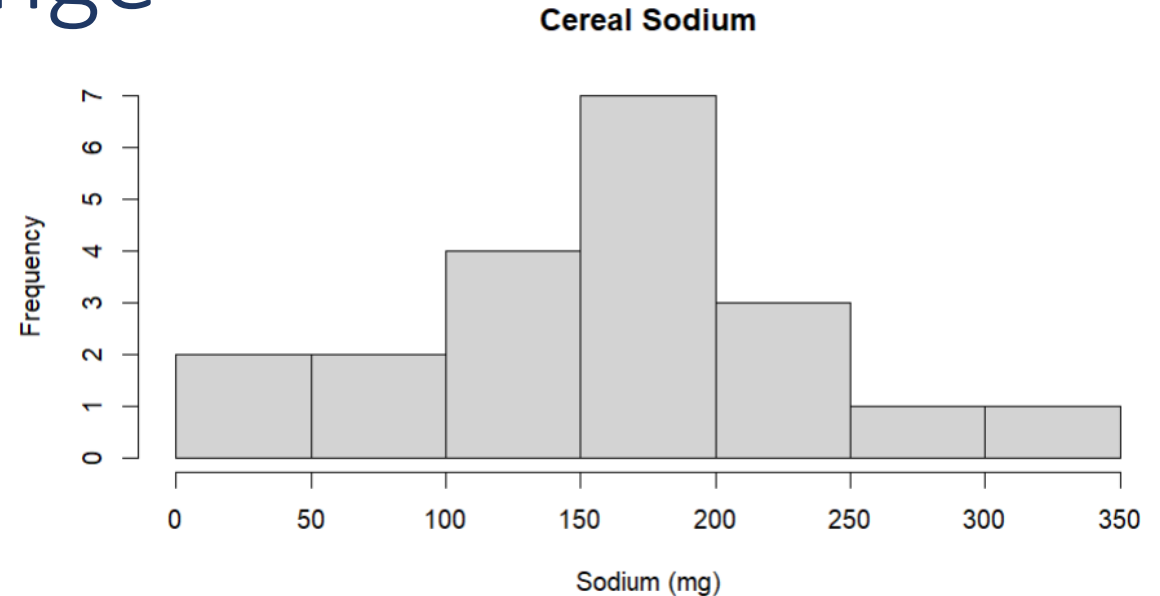


Measures of Spread: Range

- The **range** is a measure of the distance between the smallest and largest values in the data

The range can be computed with only two data points the minimum value and maximum value

- If the range of a set of data is large, then usually this indicates greater dispersion of values
- The range is severely affected by the presence of outliers
- We typically do not use the range to measure variability



Measures of Spread: Interquartile Range

- The **interquartile range (IQR)** measures the spread of the middle 50% of the observations
- It is resistant to outliers

$$IQR = Q3 - Q1$$

- The more variability the larger the value of the **IQR**
- **IQR** is a good choice for distributions that are highly skewed!

Practice: Finding quartiles and IQR

$x = 2, 4, 5, 7, 12$

Compute the range, and then the IQR

Working:

$$\text{range} = \text{max} - \text{min} = 12 - 2 = 10$$

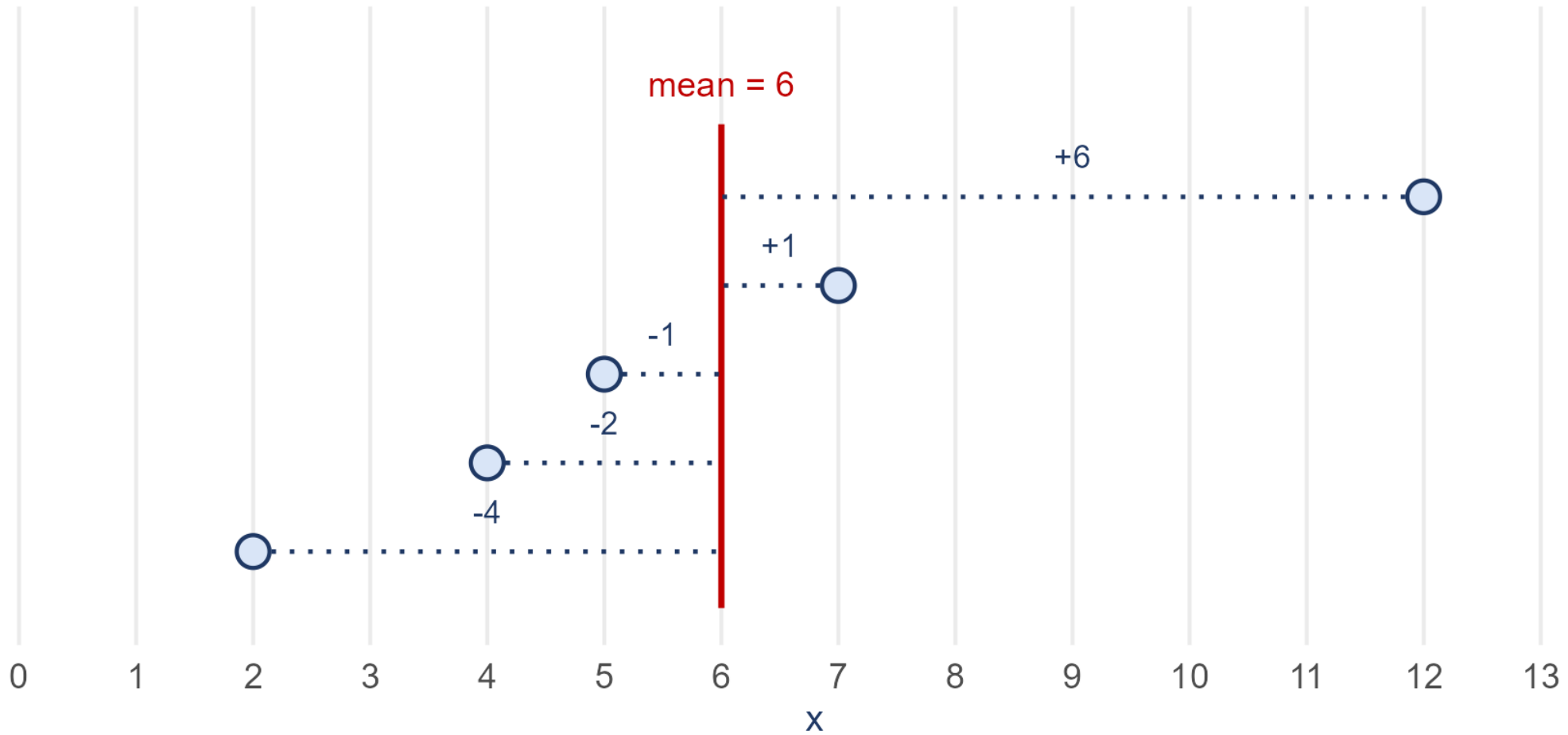
$$Q1 = 3 \quad Q3 = 9.5 \quad (\text{from the running example})$$

$$\text{IQR} = Q3 - Q1 = 9.5 - 3 = 6.5$$

The middle half spans 6.5 units; the full range spans 10.

Measures of Spread: Deviation

- A better measure of variability that uses *all* the data is based on **deviations**
- **deviations** are the distances of each value from the mean of the data:
Deviation of an observation $x_i = (x_i - \bar{x})$
- Every observation will have a deviation from the mean



Each deviation is the gap from a value to the mean. Three are negative, two positive - and they cancel exactly.

The deviations always sum to zero

- Our five cancelled exactly. That is not luck - it happens for every dataset:
 - $\Sigma(x - \text{mean}) = \Sigma x - n \cdot \text{mean}$
 - but $\text{mean} = \Sigma x / n$, so $n \cdot \text{mean} = \Sigma x$
 - **hence $\Sigma(x - \text{mean}) = \Sigma x - \Sigma x = 0$**
- Check it on our example: $-4 - 2 - 1 + 1 + 6 = 0$
- So averaging the deviations is useless - it reports 0 for tightly packed data and for wildly scattered data alike
- We have to remove the signs before averaging. That is why the next step squares them.

Measures of Spread: Variance

- The sum of all deviations is zero. $\sum_{i=1}^n (x_i - \bar{x}) = 0$
- We typically use either the **squared deviations** or their **absolute value**
Squared deviation of an observation $x_i = (x_i - \bar{x})^2$
- The **Variance** of a distribution is the average squared deviation from the mean

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

- The sum $\sum_{i=1}^n (x_i - \bar{x})^2$ is called the sum of squares

Measures of Spread: Standard Deviation

- Since the variance uses the squared deviation, we usually take its square root called the **standard deviation**

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- The standard deviation represents (roughly) the average distance of an observation from the mean
- The greater s is the greater the variability in the data is
- We denote the population parameter for the variance and standard deviation using σ for s and σ^2 for s^2

Why divide by $n - 1$?

- We divide by $n - 1$ because we have only $n - 1$ pieces of independent information for s^2
- Since the sum of the deviations must add to zero, then if we know the first $n - 1$ deviations we can always figure out the last one
- Ex.) suppose we have two data points and deviation of the first data point is $x - \bar{x} = -5$
 - Then the deviation of the second data point has to be 5 for the sum of deviations to be zero.

Try it out: computing the variance and standard deviation

- $x = 2, 4, 5, 7, 12$
- $\text{mean} = 6$
- Find the deviations, then the sum of squares, then the variance and the standard deviation

Working:

deviations -4 -2 -1 1 6

(sum 0)

squared 16 4 1 1 36

sum of squares = 58

$$s^2 = 58 / (5 - 1) = 14.5$$

$$s = \text{sqrt}(14.5) = 3.81$$

s is in the units of x ; the variance is in those units squared.

Your s is itself an estimate

- Every number today describes the one sample you happened to collect. Roll ten more times and all of them move.
- That movement from sample to sample is called **sampling variation**

	ten rolls	mean	s
our sample	1 2 3 3 4 4 4 5 6 6	3.80	1.62
sample 2	2 2 3 3 3 3 4 4 6 6	3.60	1.43
sample 3	1 2 2 3 3 4 4 4 5 6	3.40	1.51
sample 4	1 1 3 4 4 4 5 6 6 6	4.00	1.89
sample 5	1 1 2 3 4 4 6 6 6 6	3.90	2.08

How much does the mean move?

Its standard deviation across repeated samples is called the **standard error**:

$$\begin{aligned} \text{SE} &= s / \sqrt{n} \\ &= 1.619 / \sqrt{10} \\ &= 0.512 \end{aligned}$$

Bigger n, smaller SE.

Not on Exam 1 - this is the idea behind margin of error, which we pick up in a few weeks.

Which measures do you report?

- A center on its own does not describe a distribution - it needs a spread beside it, and the two come in matched pairs
 - the mean and the standard deviation – both use every observation, so both are pulled by an outlier
 - the median and the IQR – both are positions in the ordered data, so both are resistant
- A measure is **resistant** when a single extreme observation barely moves it
- Roughly symmetric, no outliers: report the mean and the standard deviation
- Skewed, or outliers present: report the median and the IQR
- Do not mix the pair. A mean next to an IQR describes two different things and answers neither question well.

The five number summary

- **Five numbers** are enough to sketch a whole distribution

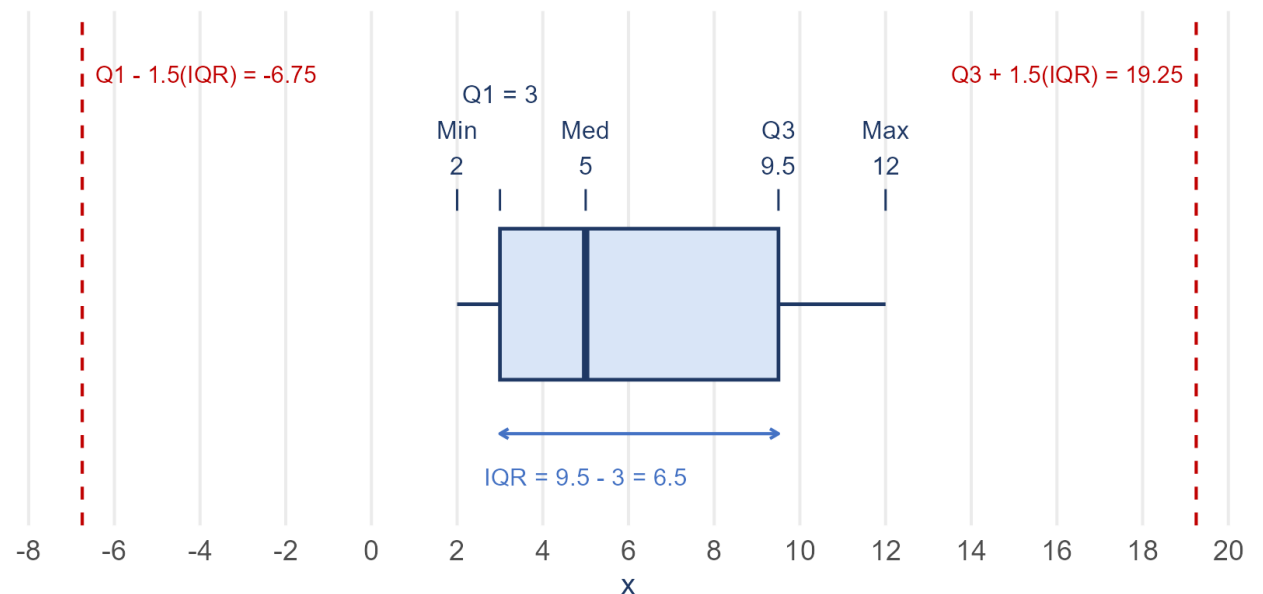
- **Minimum Q1 Median Q3 Maximum**

- **For $x = 2, 4, 5, 7, 12$: 2 3 5 9.5 12**

- the median gives the center
- Q1 and Q3 bracket the middle half, and their difference is the IQR
- the minimum and maximum give the full extent
- These five numbers are exactly what a boxplot draws

The Boxplot (Box and Whisker Plot): A five number summary

- Pros:
 - good for describing shape and location
 - Can be used to identify outliers
 - Length of whiskers indicates skew
 - Good for comparing two distributions or across categories
- Cons:
 - does not show certain features like mounds, or gaps as well as a histogram



Identifying outliers: the 1.5 x IQR rule

- We need a rule for "extreme" that does not depend on eyeballing the plot
- The **fences** sit an extra one and a half IQRs beyond the box:
 - $Q1 - 1.5(IQR)$ and $Q3 + 1.5(IQR)$
- Any observation outside a fence is an **outlier**, and gets plotted as its own point
- The whiskers then stop at the most extreme observation INSIDE the fences - not at the minimum and maximum
- The fences themselves are never drawn on the plot
- Our example: $3 - 1.5(6.5) = -6.75$ and $9.5 + 1.5(6.5) = 19.25$, so nothing is flagged and the whiskers reach 2 and 12
- Being flagged does not make an observation wrong. It makes it worth a second look.
- Careful: R's `boxplot()` would draw this box at 4 and 7, not 3 and 9.5 - it uses Tukey's hinges, the third row of the table back on slide 4

Ex. Construct a Boxplot

- Consider the following data which come from 20 rolls of a six-sided die

• Data =

lower half	middle	upper half
1, 1, 1, 2, 2, 2, 2, 2	2, 3	3, 3, 3, 3, 4, 5, 5, 6, 6, 6

$$Q2 = \text{median} = \frac{2+3}{2} = 2.5$$

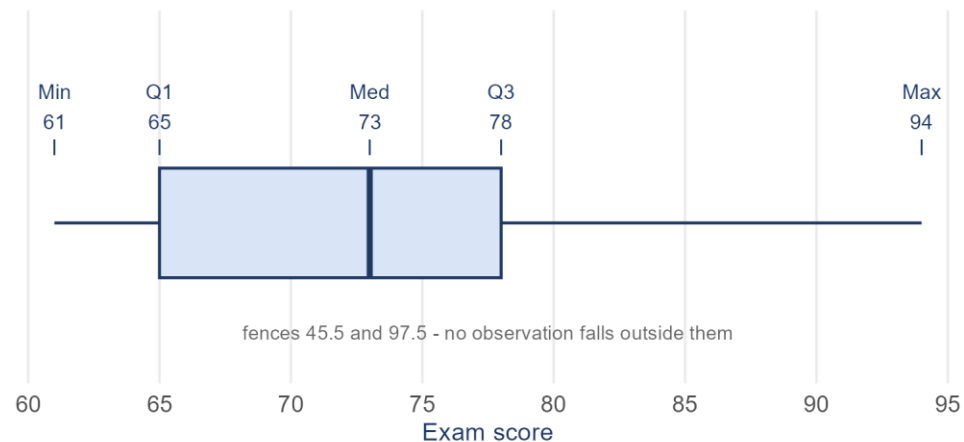
$$Q1 = \text{median lower half} = \frac{2+2}{2} = 2$$

$$Q3 = \text{median upper half} = \frac{4+5}{2} = 4.5$$

Practice: Construct a Boxplot

Construct the boxplot for student exam scores

Exam Scores 61,61,65,65,66,68,69,73,74,75,76,78,79,90,94



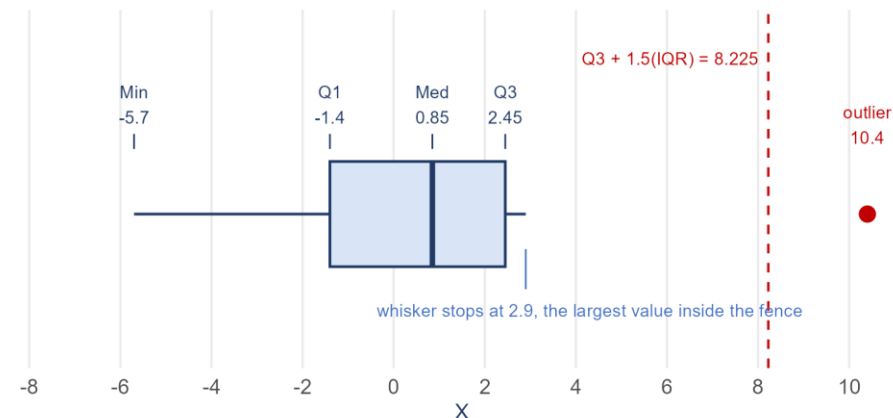
Ex2. Construct a Boxplot

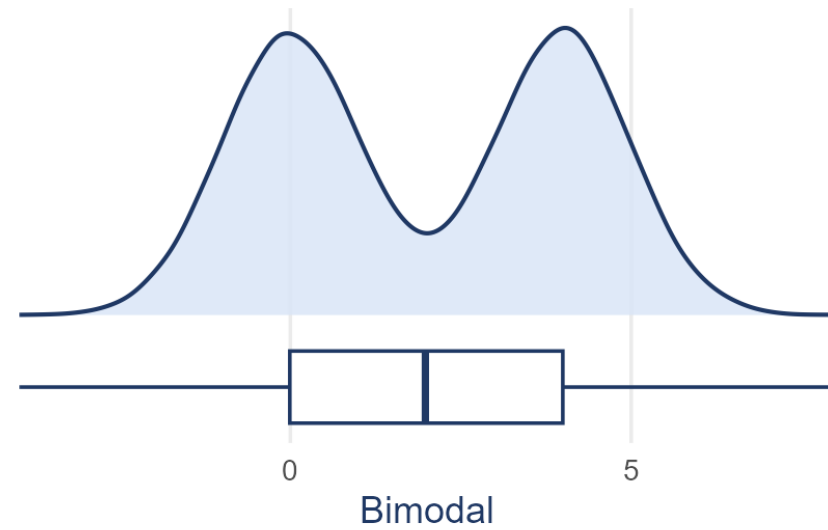
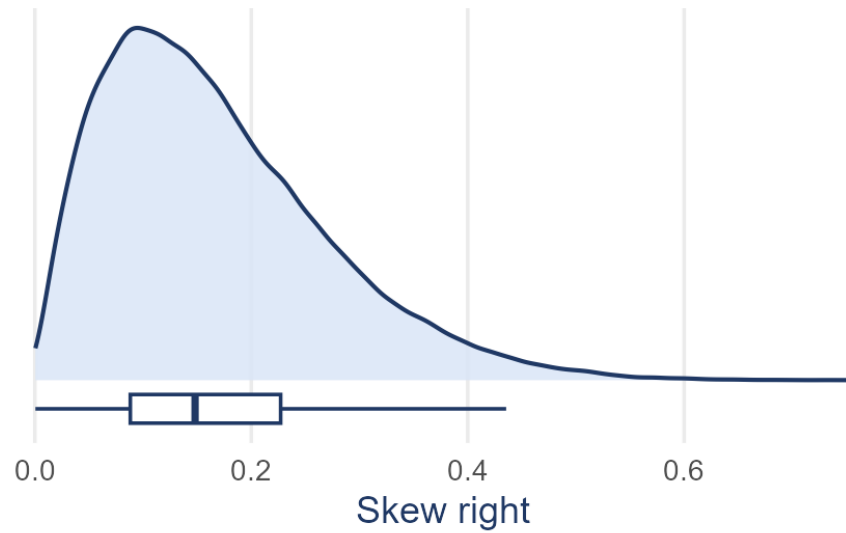
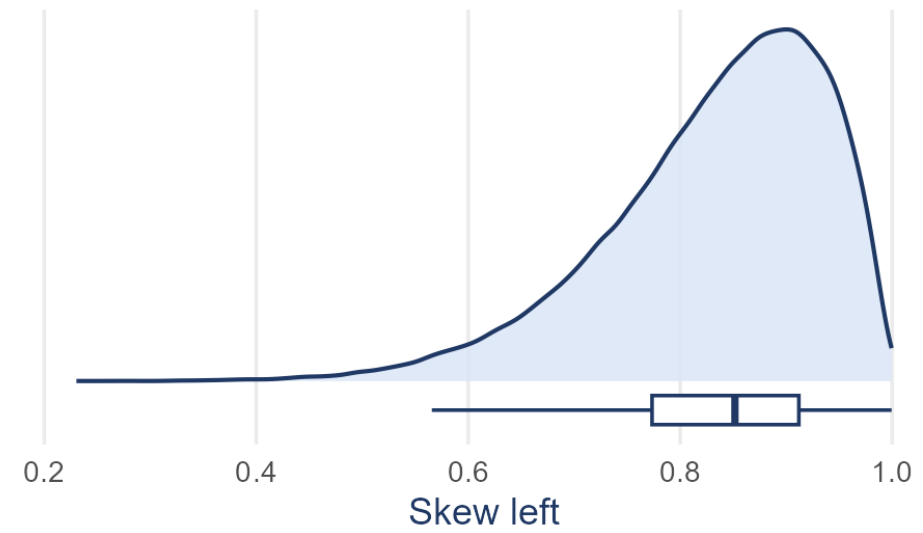
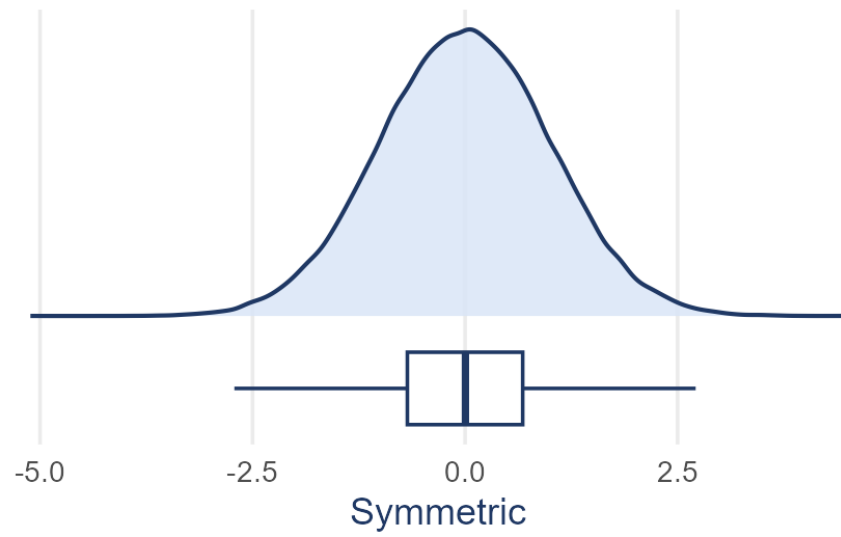
Consider the following 12 observations of a quantitative variable X

$X = \{-5.7, -2.6, -1.5, -1.3, -0.4, 0.2, 1.5, 2.2, 2.3, 2.6, 2.9, 10.4\}$

Compute the 5 number summary and draw a boxplot

Compare the IQR with the range





Whisker lengths show skew. But the bimodal panel has the same boxplot as a symmetric one - a boxplot cannot show peaks or gaps, so read it next to a histogram.